Assignment : 04

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Time Complexity :

1) $n = 4$

```
for(int i = 1 ; i ≤ n ; i++){
    for (int j = 1 ; j ≤ n ; j++){
        print (*) ;
    }
    S.opln()
}
```

```
* * * *
* * * *
* * * *
* * * *
```

$i = 1$

$j = 1, 1 \leq 4$ — ①

$= 2 \leq 4$ (j++) — $n+1$

$= 3 \leq 4$ (j++) — 1

$= 4 \leq 4$ (j++) — 1

$= 5 \leq 4$ ✗ (j++) — 1

$i = 2$

$j = 1 \leq 4$

$= 2 \leq 4$

$3 \leq 4$

$4 \leq 4$

$5 \leq 4$ ✗

$i = 1$, $j = 5$ | $i = n$, $j = n+1$

i depends on n.

$$TC = O\left(\frac{n(n+1)}{2}\right)$$
$$= O\left(\frac{n^2+n}{2}\right)$$
$$= O\left(\frac{n^2}{2} + \frac{n}{2}\right)$$
$$= O\left(\frac{n^2}{2}\right) = \boxed{O(n^2)}$$

2)

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4.
```

Count = ✗ 2.

$i = 1 \leq 4$ — ①

$j = 1 \leq 4$ — $(n+1)$

$2 \leq 4$ — ①

$3 \leq 4$ — ①

$4 \leq 4$ — ①

$5 \leq 4$ ✗ — ①

$i = 2 \leq 4$

$$\boxed{T.C = O(n^2)}$$

3) 1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4.

```
count = 1
for (int i = 1; i ≤ n; i++) {
    for (int j = 1; j ≤ n; j++) {
        print (count);
        count++;
    }
}
```

$$\boxed{\begin{array}{l} TC = O(n^2) \\ SC = O(1) \end{array}}$$

4)
```
* * * * *
*       *
*       *
*       *
* * * * *
```

```
n = 5
for (int i = 1; i ≤ n; i++) {
    for (int j = 1; j ≤ n; j++) {
        if (i == 1 || i == 5 || j == 1 || j == 5)
            print (*);
        } else {
            print ();
        }
    }
```

i = 1 (1)
j = 1 ≤ 5 (n+1) → *
2 ≤ 5 1 → —
3 ≤ 5 1 → —
4 ≤ 5 1 → —
5 ≤ 5 1 → *
6 ≤ 5 1 ✗

i = 2.
j = 1 ≤ 5 → *
2 ≤ 5 → —
3 ≤ 5 → —
4 ≤ 5 → —
5 ≤ 5 → *
6 ≤ 5 ✗

$$TC = O\left(n + \left(\frac{n+1}{2}\right)\right)$$
$$= O\left(n^2 + \frac{n}{2}\right)$$
$$= O\left(\frac{n^2}{2} + \frac{n}{2}\right)$$
$$= O\left(\frac{n^2}{2}\right)$$

$$\boxed{TC = O(n^2)}$$

n = 5

5)

```
*       *
  *   *
    *
  *   *
*       *
```

```
for (int i=1; i≤n ; i++) {
    for (int j=1, j≤n; j++) {
        if ( i==j || i+j=n+1){
            print ("*");
        } else {
            print (" ");
        }
    }
}
```

$\boxed{T.C = O(n^2)}$

n=5

6)

```
*
* *
* * *
* * * *
* * * * *
```

```
for (int i=1; i≤n ; i++) {
    for (int j=1 ; j≤i ; j++) {
        print ("*");
    }
    sopln()
}
```

$\boxed{T.C = O(n^2)}$

n=5

7)

```
* * * * *
* * * *
* * *
* *
*
```

```
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; j≤n-i+1, j++){
        print ("*");
    }
    popln ()
}
```

i=1

j=1 ≤ n+1−i = 5+1−i = 6−1 = 5

=2 ≤ 5

=3 ≤ 5

=4 ≤ 5

=5 ≤ 5

=6 ≤ 5 ×

$\boxed{TC = O(n^2)}$

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$n = 4.$

8)

```
* # * #
# * # *
* # * #
# * # *
```

```
for (int i=1 ; i≤n ; i++){
  for (int j=1 ; j≤n ; j++){
    if ( i+j % 2 == 0 ){
      print ( * ) ;
    } else {
      print (#) ;
    }
```

$\boxed{T.C = O(n^2)}$

Q.1)

steps

1.
2.
3.
4.
5.